



HAI KUANG PACKING INDUSTRIES SDN. BHD. (266530-W)

海光包裝工業有限公司

No. 12 & 14, Jalan 5/10B, Spring Crest Industrial Park, 68100 Batu Caves, Kuala Lumpur, Malaysia.

Tel: +603-6187 5720, 6187 4177 Fax: +603-6189 9530 Email: hkpack@haikuang.com

TECHNICAL DATA SHEET

Section 1. IDENTIFICATION OF PRODUCT

Product name: SPE Metal Ink
Product code: SPE-(e.g: 1481, 1482, 1490, 2333)
Intended use: Silkscreen printing ink

Section 2. INFORMATION OF INGREDIENTS

Organic and inorganic pigment dispersion in synthetic polymers with organic solvents and additives as required.

Hazardous ingredients	Conc. Range %	Symbol	TLV
Glycol ethers	$\geq 20 \leq 35$	XN	25 ppm (skin) TWA (ACGIH)
Glycol esters	$\geq 5 \leq 11$	F	5 ppm (skin) TWA (ACGIH)
Aromatic hydrocarbon	$\geq 5 \leq 14$	XN	25 ppm TWA (ACGIH)

Section 3. HAZARDS IDENTIFICATION

Harmful if swallowed. Exposure to vapor at high concentrations may have the following effect - nausea, headache, dizziness, symptoms similar to alcohol intoxication. Eye contamination may result in conjunctiva irritation and possible corneal damage. Prolonged or repeated skin exposure to product containing solvents may result in dermatitis unless good personal hygiene is observed.

Health studies have shown that many petroleum hydrocarbons pose potential human health risk which may vary from person to person. As a precaution, exposure to liquids, vapors, mist or fumes should be minimized.

Section 4. FIRST AID MEASURES

General - In all cases of doubt or when symptoms persist, seek immediate medical advice. Never give anything by mouth to an unconscious person.

Inhalation - Move to fresh air in cases of accidental inhalation of vapors.

Skin contact - Take off all contaminated clothings. Wash off immediately with soap and plenty of water.

Eye contact - Rinse immediately with plenty of water and seek medical advice.

Ingestion - Seek immediate medical attention. Keep a rest. Do not induce vomiting.

Section 5.**FIRE FIGHTING MEASURES**

Extinguishing media – Foam, carbon dioxide, dry powder sand, water spray mist.

NO TO BE USED – High volume water jet.

Specific hazards – Burning produces irritant fumes, hazardous combustion products. Closed containers may explode at elevated temperature. Reduce pressure and risk of auto ignition and explosion by cooling containers with water spray.

Special protective equipment for fire-fighters – Wear self-contained breathing apparatus.

Section 6.**ACCIDENTAL RELEASE MEASURES**

Personal precautions – In cases of vapor formation, use respirator.

Environmental precautions – Do not allowed to enter drains or water courses. Contain and collect spillage with inert absorbent material (e.g. Sand, earth) and place in suitable container for disposal in accordance with waste regulations.

Further advice – Clean contaminated surface thoroughly.

Section 7.**HANDLING & STORAGE****Handling**

Personal protection – avoid skin and eye contact. Good housekeeping and well ventilation will minimize risks.

Storage

Keep away from sources of ignition. No smoking.

Store in well ventilated area and away from direct heat and sunlight.

Keep containers closed and upright. Always ensure that product after opened or partially used, lids are well resealed or closed. Precaution to be taken against possibility of static discharge.

Section 8.**EXPOSURE CONTROLS/ PERSONAL PROTECTION**

Engineering measures – All areas should be well ventilated. Use local exhaust ventilation and good general extraction. (Refer to SECTION 2 for exposure controls)

Hygiene measures – Handle in accordance with good industrial hygiene and safety practice.

Respiratory protection – In cases of insufficient ventilation, wear suitable respiratory equipment.

Eye protection – Safety glasses.

Skin protection – Chemical resistant synthetic overalls or coveralls.

Hand protection – Chemical or solvent resistant gloves.

Section 9.**PHYSICAL & CHEMICAL PROPERTIES**

Physical state

Color paste

Flash point – Tag closed cup

43

Explosion limit in air (volume %)

Lower : 1.0 Upper : 6.5

Vapor pressure (Air = 1)

Heavier than air

Boiling range

150 - 230

Solubility in water

Immiscible

Volatile

>37%, <60% by weight (varies with color)

Section 10.**STABILITY & REACTIVITY**

Stability

Product is stable under normal ambient conditions

Material to avoid

Strong oxidizing agents

Section 11.**TOXICOLOGICAL INFORMATION**

Data is not determined and available on the product itself.

Skin - repeated or prolonged contact

Eye contamination

Health

: Removal of natural fats from skin, mild irritation.
: Irritation and reversible local damage.
: Exposure to organic solvent vapors of mucous membrane and respiratory system. Symptoms - headache, dizziness, fatigue, muscular weakness, drowsiness and in extreme cases, unconsciousness.
Ingestion may cause irritation, nausea and dizziness.

Section 12.**ECOLOGICAL INFORMATION**

Do not allow product to enter drains and water courses or be deposited where it can affect ground or surface water.

Section 13.**DISPOSAL CONSIDERATIONS**

Waste from residues / unused product :

Can be incinerated when in compliance with local regulations.

Contaminated packaging :

Empty containers should be disposal of in accordance with local regulation.

Section 14.**TRANSPORT INFORMATION**

Un No

Hazchem code

Designation

Packaging Group

Class

Not available

3 YE

Flammable liquid

III

3.3

Section 15.**REGULATORY INFORMATION**

This product is classified and labeled for supply in accordance with the Chemicals (Hazard Information & Packaging) Regulations as follow : -

Classification

F - Flammable

Risk phrases

10 - Flammable

22 - Harmful if swallowed

Safety phrases

16 - Keep away from sources of ignition - " NO smoking"

21 - When using do not smoke

29 - Do not empty into drains

Section 16.**OTHER INFORMATION**

The information is based on our present state of knowledge. It provides guidance on health, safety and environmental aspects of product. It does not constitute a guarantee for product characteristics and technical performance. As the specific conditions of use of the product is outside of manufacture's / supplier's control, the user is responsible for ensuring that the requirements of relevant legislation is complied with.

Issuing date

: 26th September 04

Revision date

: Not applicable